

DEMONSTRATION OF PROPOSED FEATURES

OptiCrop: Smart Agricultural Production Optimization Engine

Date	03 July 2026
Team ID	SWTID-2026-7253
Team Size	5
Team Leader	Bommidi Deepika
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Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated[cite: 38]. It serves as evidence of the team's ability to deliver on the proposed solution[cite: 38].

Feature Ingestion and Demonstration Matrix

S.No	Feature Name	Description	Status	Demonstrated (Yes / No)	Remarks
1	User Management	Secure user profiles and login access portal built with cryptographic encryption mechanisms[cite: 38].	Implemented[cite: 38]	Yes[cite: 38]	Functional as per PEP software standards[cite: 38].
2	Soil & Climate Form	Interactive interface form capturing regional coordinates and specific target land metrics[cite: 38].	Implemented[cite: 38]	Yes[cite: 38]	Includes complete real-time client side validation[cite: 38].
3	Preprocessing Core	Handles environmental vector scaling, dataset noise extraction, and matrix array transformations[cite: 38].	Implemented[cite: 38]	Yes[cite: 38]	Built via optimized Pandas and Scikit-learn systems[cite: 38].
4	Crop Prediction Engine	Processes parameter profiles through calibrated pipelines for high-precision multi-class recommendations[cite: 38].	Implemented[cite: 38]	Yes[cite: 38]	Generates suitability matching indices and confidence metrics[cite: 38].

S.No	Feature Name	Description	Status	Demonstrated (Yes / No)	Remarks
5	Database Layer	Secure data tier handling transaction persistence and history logging for farmers and researchers[cite: 38].	Implemented[cite: 38]	Yes[cite: 38]	Using relational MySQL clustered architecture setups[cite: 38].
6	Analytics Dashboard	Administrative interface summarizing regional soil characteristics, tracking counts, and exporting metrics[cite: 38].	Implemented[cite: 38]	Yes[cite: 38]	Provides dynamic analytics and custom reporting options[cite: 38].

Feature Implementation Summary

Metric Category	Value
Total Features Proposed	6[cite: 38]
Total Features Implemented	6[cite: 38]
Total Features Demonstrated	6[cite: 38]
Overall Implementation Rate (%)	100%[cite: 38]